Product Requirements Document

(PRD)

Falcon Academic Management System

(FAMS)

Enterprise ERP System for Multi-Campus Education Management

Client Organization	Falcon College
System Name	Falcon Academic Management System (FAMS)
Document Type	Product Requirements Document (PRD) — Version 1.0
Document Status	Final Draft — Awaiting Stakeholder Approval
Deployment Model	Web-Based, Cloud-Enabled, Centralized
Scale	31 Campuses — Enterprise Multi-Campus ERP
Prepared By	Mehboob.mov
Classification	Confidential

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY

This Product Requirements Document (PRD) defines the complete specification for the Falcon Academic Management System (FAMS), a purpose-built enterprise ERP platform commissioned by Falcon College to unify the administrative, academic, financial, and human capital operations of its 31-campus network under a single, centralized digital ecosystem. The document serves as the authoritative reference for product development, stakeholder alignment, quality assurance, and post-launch governance.

Falcon College currently operates across 31 geographically distributed campuses, each running partially independent administrative workflows on a combination of legacy software, paper-based processes, and disconnected spreadsheet systems. This fragmentation creates significant operational overhead, data inconsistencies, reporting delays, and a degraded service experience for staff, students, and parents. FAMS is designed to eliminate these inefficiencies by providing a unified, cloud-enabled, web-based platform accessible to all authorized stakeholders through role-differentiated portals.

The system encompasses eight core functional modules: Customer Relationship Management (CRM), Admissions, Procurement and Vendor Management, Academic Operations, Results and Reporting, Accounting and Finance, Human Resource Management (HRM), and Assets and Inventory Management. Overlaying all modules are cross-cutting enterprise features including real-time analytics dashboards, AI-powered chatbot support, tablet-based attendance capture, e-learning integration, and fully paperless transactional workflows.

Total Campuses	31
Deployment Model	Centralized Cloud-Enabled Web Application
Primary Stakeholders	Board of Governors, Campus Principals, Administrative Staff, Teaching Staff, Students, Parents
Core Modules	8 Functional Modules + Cross-Cutting Platform Layer
Document Version	1.0 — Final Draft
Classification	Confidential

2. PRODUCT VISION

The vision for FAMS is to establish Falcon College as a digitally mature, data-driven educational institution capable of delivering consistent, high-quality services across all 31 campuses through a single, integrated technology platform. FAMS is conceived not merely as a software tool but as the central nervous system of the institution — connecting every operational domain, surfacing actionable intelligence in real time, and enabling leadership to make informed decisions grounded in accurate, institution-wide data.

The platform embodies four foundational design principles: (i) *Centralization* — all campuses operate from a single source of truth, eliminating data silos; (ii) *Transparency* — every transaction and approval is logged, traceable, and auditable; (iii) *Accessibility* — role-appropriate portals ensure every stakeholder can access relevant information securely; and (iv) *Scalability* — the architecture accommodates institutional growth without structural re-engineering.

The long-term product vision positions FAMS as an AI-augmented institutional intelligence platform where predictive analytics inform admissions strategy, machine learning models identify at-risk performance patterns, and automated workflows eliminate manual overhead across all departments.

3. OBJECTIVES

3.1 Operational Objectives

- Consolidate all campus-level administrative operations onto a single platform, eliminating the use of standalone legacy systems and paper-based workflows.
- Reduce administrative processing time for key transactions (admissions, fee collection, result publication, payroll) by a minimum of 60 percent.
- Enable real-time attendance capture across all campuses using tablet-based hardware, replacing manual register systems.
- Automate end-to-end fee management from invoice generation to receipt issuance and defaulter identification, with zero manual reconciliation.
- Provide principals and administrative officers with role-specific dashboards surfacing live KPIs relevant to their area of responsibility.

3.2 Strategic Objectives

- Create a unified institutional data repository supporting cross-campus benchmarking, performance analysis, and evidence-based policy decisions.
- Improve the service experience for students and parents through self-service portals providing 24/7 access to academic records, fee statements, and communications.
- Enable the Board of Governors to monitor institution-wide KPIs through a consolidated executive dashboard.
- Establish the data infrastructure required to support future AI-driven analytics and predictive modeling capabilities.
- Transition Falcon College to a fully paperless operational model within 18 months of go-live.

3.3 Technology Objectives

- Deliver a cloud-enabled, centrally hosted architecture with 99.9 percent uptime availability across all campuses and user roles.
- Implement enterprise-grade security including role-based access control, data encryption at rest and in transit, and comprehensive audit logging.
- Ensure real-time data synchronization across all 31 campuses with maximum propagation latency under five seconds for critical transactional data.
- Integrate an AI-powered chatbot capable of resolving Tier-1 support queries for all user roles without human intervention.
- Build the platform on an API-first architecture to facilitate future integrations with third-party services, government data systems, and e-learning platforms.

4. SCOPE

4.1 In Scope

The following functional domains, features, and deliverables are explicitly included within the scope of the FAMS v1.0 release.

Core ERP Modules	All eight functional modules as defined in Section 7.
Multi-Campus Management	Centralized data management for all 31 campuses with campus-level configuration and reporting.
Role-Based Portals	Distinct portals for System Administrator, Principal, Teacher, Student, Parent, Accountant, HR Officer, and Procurement Officer.
Real-Time Dashboards	KPI dashboards with live graphs, trend analytics, and exportable reports for all management roles.
Tablet Attendance System	Tablet-optimized interface for attendance capture in classrooms and examination halls.
Online Services	Web-accessible services for admissions, fee payment, result viewing, and institutional communications.
AI Chatbot	Integrated conversational AI assistant providing Tier-1 support across all user portals.
E-Learning Integration	Integration layer connecting FAMS to a designated external LMS for content delivery and progress tracking.
Notifications Engine	Automated SMS, email, and in-app notifications for defined trigger events.
Data Migration	Structured migration of existing student, staff, and financial data from legacy systems.
System Integrations	REST API integrations with payment gateways and SMS gateways.
Security Framework	Role-based access control, AES-256 encryption, audit trails, and multi-factor authentication.
Training & Documentation	User manuals, administrator guides, and onboarding training materials for all user roles.

4.2 Out of Scope

- Native mobile application development (iOS / Android) — a mobile-responsive web interface will be delivered instead.
- Biometric hardware provisioning, installation, and maintenance — hardware is the responsibility of the client's IT team.
- Custom Learning Management System (LMS) development — FAMS will integrate with an externally provisioned LMS.
- Government examination board integration — planned for Phase 2 roadmap.
- Alumni management and post-graduation tracking modules.
- Library management, hostel/accommodation, and transportation management.
- Third-party ERP system data synchronization beyond the agreed migration scope.
- Custom BI tooling beyond the built-in dashboards.

5. SYSTEM OVERVIEW

FAMS is architected as a cloud-enabled, multi-tier web application built on a microservices-oriented backend, a centralized relational database, and a responsive single-page application (SPA) frontend. The system is accessible via any modern web browser on desktops, laptops, and tablet devices, with the attendance module optimized for tablet-first interaction as a progressive web application (PWA) supporting offline operation.

The platform operates on a centralized database model — all 31 campuses write to and read from a single logical data store. Campus-level data isolation is enforced at the application layer through a tenant-aware data access model using a campus_id discriminator, ensuring administrators can only access data pertaining to their assigned campus unless explicitly granted cross-campus visibility.

Presentation	Web Portal (SPA)	React.js — Responsive, role-based UI
Presentation	Tablet Attendance UI	Progressive Web App (PWA) — Offline-capable
API Gateway	Centralized Gateway	RESTful API, JWT auth, rate limiting
Application	Core Business Logic	Node.js / Python microservices
Application	AI Chatbot Engine	NLP-based conversational AI
Application	Notifications Service	Async event-driven message broker
Data	Primary Database	PostgreSQL — row-level security (RLS)
Data	Cache Layer	Redis — sessions and hot data
Data	File Storage	Cloud object storage (S3-compatible)
Infrastructure	Cloud Hosting	AWS / Azure — multi-AZ deployment
Integration	Payment Gateway	REST API — third-party PSP
Integration	SMS / Email Gateway	Transactional messaging API
Integration	LMS Bridge	API SSO to external LMS

6. USER ROLES AND PERSONAS

FAMS implements a strict role-based access control (RBAC) model. Each user is assigned one or more roles at account provisioning. Roles determine portal access, module visibility, data scope, and permitted actions. Table I summarizes the defined roles, their organizational context, and primary responsibilities.

System Administrator	IT / RaideIT ops team	Full system access, all campuses. User provisioning, system configuration, security, integrations.
Executive / Board	Board of Governors, Group Director	Read-only institution-wide KPI dashboards. Financial summaries, enrollment trends, academic performance across all campuses.
Principal	Campus Principal / Head	Full operational access scoped to assigned campus: staff, students, timetables, results, financials.
Academic Coordinator	HOD, Exam Controller	Academic operations: timetable management, examination scheduling, result processing, attendance reporting.
Teacher	Teaching staff	Class timetables, attendance marking, gradebook entry, assigned student profiles.
Accountant	Campus finance team	Fee management, payment records, payroll processing, vendor invoices, financial reports.
HR Officer	Human resources team	Staff profiles, recruitment, leave management, attendance records, HR reporting.
Student	Enrolled students	Self-service: timetable, attendance, exam schedule, results, fee statements, e-learning, communications.
Parent / Guardian	Parents and legal guardians	Child's attendance, academic performance, fee status, examination results. Notification receipt and query submission.
Procurement Officer	Campus procurement team	Purchase requisitions, vendor records, purchase orders, goods receipts, vendor invoices.

Table I. User Roles and Access Summary

7. FUNCTIONAL REQUIREMENTS

This section documents the detailed functional requirements for each of the eight core modules and the cross-cutting platform features. Requirements are presented in terms of system capabilities, user-facing features, and operational workflows.

7.1 CRM — Student and Parent Management

The CRM module serves as the central repository for all student and parent profiles, managing the complete lifecycle from initial inquiry through active enrollment to alumni status. It provides a 360-degree view of each student, aggregating data from academic, financial, attendance, and communication modules.

FR-CRM-01	Student Profile Management	Create, update, and archive comprehensive student profiles including personal information, emergency contacts, medical notes, and academic history.
FR-CRM-02	Parent/Guardian Profiles	Maintain linked parent profiles with contact preferences, communication history, and portal access credentials.
FR-CRM-03	Student Lifecycle Tracking	Track status transitions: Prospect → Applicant → Enrolled → Active → Graduated / Withdrawn, with timestamps and officer attribution.
FR-CRM-04	360-Degree Student View	Unified interface aggregating attendance, academic performance, fee status, behavioral notes, and communication history.
FR-CRM-05	Communication Log	Record all institutional communications (SMS, email, in-app) with timestamps and initiating officer attribution.
FR-CRM-06	Segmentation and Filtering	Advanced search allowing filtered views by campus, class, performance band, fee status, or custom criteria.
FR-CRM-07	Document Management	Attach and manage digital copies of admission forms, birth certificates, transfer certificates against student profiles.
FR-CRM-08	Inquiry Management	Capture and track prospective inquiries from all channels with follow-up task assignment and status tracking.
FR-CRM-09	Family Linkage	Link multiple student profiles within the same family for consolidated parent-facing views and family-level fee management.

7.2 Admissions and Enrollment Management

The Admissions module manages the end-to-end enrollment pipeline from initial inquiry through application submission, merit evaluation, offer issuance, fee deposit, and final enrollment confirmation.

FR-ADM-01	Online Application Portal	Applicants complete and submit enrollment applications via web portal with document upload capability.
FR-ADM-02	Application Pipeline Dashboard	Visual funnel view of all applications by stage: Inquiry, Applied, Under Review, Offered, Enrolled, Declined.
FR-ADM-03	Merit List Generation	Automated merit list computation based on configurable criteria (marks, entry test, quotas). PDF export.

FR-ADM-04	Seat Availability Management	Real-time seat tracking per campus, program, and section. Automatic wait-list management.
FR-ADM-05	Offer Letter Generation	Automated generation and email delivery of offer letters with enrollment deadline and fee instructions.
FR-ADM-06	Enrollment Confirmation	Enrollment confirmed upon fee deposit receipt, triggering profile activation and portal credential issuance.
FR-ADM-07	Transfer Applications	Manage inter-campus transfer requests with approvals, document verification, and academic record migration.
FR-ADM-08	Bulk Import	Support bulk enrollment data import via structured CSV/Excel templates for large cohort onboarding.
FR-ADM-09	Admissions Analytics	Conversion funnel analytics: inquiry-to-application rate, application-to-enrollment rate, campus-wise trends.

7.3 Procurement and Vendor Management

The Procurement module governs the institutional purchasing lifecycle from requisition to payment, ensuring compliance with approval hierarchies, budget controls, and vendor contractual obligations.

FR-PRC-01	Vendor Registry	Centralized vendor master with profiles, payment terms, approved categories, and performance ratings.
FR-PRC-02	Purchase Requisition	Digital requisition workflow with multi-level approval (HOD → Principal → Finance) based on value thresholds.
FR-PRC-03	Purchase Order Management	Automated PO generation from approved requisitions with version control and vendor delivery.
FR-PRC-04	Goods Receipt Note (GRN)	Digital GRN creation upon delivery with quantity verification against PO and inventory update trigger.
FR-PRC-05	Vendor Invoice Processing	Three-way match (PO, GRN, Invoice) verification before invoice approval.
FR-PRC-06	Budget Integration	Real-time budget utilization checks against departmental budgets before requisition approval.
FR-PRC-07	Vendor Performance Tracking	Scorecard tracking of on-time delivery, quality compliance, and invoice accuracy.
FR-PRC-08	Contract Management	Store and track vendor contracts with expiry alerts and renewal workflow.
FR-PRC-09	Procurement Analytics	Spend analysis by campus, category, vendor, and time period. Supplier concentration risk reporting.

7.4 Academic Operations

The Academic Operations module is the operational core of FAMS, managing timetabling, attendance, and examination administration across all campuses.

7.4.1 Timetable Management

FR-TT-01	Automated Timetable Generation	Configurable engine generating weekly schedules based on teacher availability, subject load, room capacity, and section assignment.
FR-TT-02	Conflict Detection	Real-time prevention of teacher double-bookings, room conflicts, and student schedule overlaps.
FR-TT-03	Substitute Teacher Assignment	Workflow for assigning substitute teachers to absent teacher slots with one-click student notification.
FR-TT-04	Timetable Publishing	Campus-specific timetables published to teacher and student portals with printable PDF export.
FR-TT-05	Session Scheduling	Support for term/semester-based academic calendars with holiday and examination period blocking.

7.4.2 Attendance Management

FR-ATT-01	Tablet-Based Attendance Capture	Teachers mark student attendance via tablet-optimized interface. No paper registers required.
FR-ATT-02	Real-Time Attendance Sync	Records synchronize to central database within seconds, immediately visible to parents and administrators.
FR-ATT-03	Staff Attendance Tracking	Daily staff check-in/check-out with late arrival and early departure flagging.
FR-ATT-04	Absence Alerts	Automated SMS/email notifications to parents upon student absence within the same session.
FR-ATT-05	Attendance Analytics	Daily, weekly, and monthly reports by student, class, section, subject, and campus.
FR-ATT-06	Leave Application Workflow	Students submit leave requests via portal. Teachers and principals approve with documentation attachment.
FR-ATT-07	Attendance-Based Eligibility	Configurable attendance thresholds for examination eligibility with automated eligibility reports.

7.4.3 Examination Management

FR-EXM-01	Examination Scheduling	Create and publish examination timetables by term, subject, and section with room allocation.
FR-EXM-02	Admit Card Generation	Automated admit card generation and distribution via student portal with eligibility enforcement.
FR-EXM-03	Seating Plan Management	Automated seating plan generation for examination halls based on enrollment and hall capacity.
FR-EXM-04	Invigilator Assignment	Assign invigilators per examination session with conflict-of-interest checks against teaching assignments.
FR-EXM-05	Answer Script Tracking	Log and track answer script bundles from collection to marking to return.
FR-EXM-06	Examination Fee Management	Integrated with Finance module for examination fee invoicing and payment verification.

7.5 Results and Reporting

The Results module manages the complete academic assessment lifecycle from marks entry through result computation, grade card generation, and institutional performance reporting.

FR-RES-01	Marks Entry Interface	Teacher-facing marks entry by subject, assessment type, and student. Bulk import via structured templates.
FR-RES-02	Configurable Grading System	Support for percentage-based, GPA, and custom grading scales configured per program or campus.
FR-RES-03	Automated Result Computation	System computes final grades, aggregate scores, division classification, and pass/fail status.
FR-RES-04	Grade Card Generation	Automated production of individual grade cards in institutional format, available on student and parent portals.
FR-RES-05	Result Publishing Workflow	Multi-stage approval: Teacher submits → Coordinator verifies → Principal approves → Published.
FR-RES-06	Academic Progress Tracking	Longitudinal performance records across terms and academic years with trend visualization.
FR-RES-07	Class/Section Rank Computation	Automatic ranking of students within class and section based on aggregate scores.
FR-RES-08	Institutional Performance Reports	Cross-campus analytics: subject pass rates, grade distributions, teacher effectiveness proxies.
FR-RES-09	At-Risk Student Reports	Automated identification of students below passing threshold for timely intervention.
FR-RES-10	Result Notifications	Parents and students notified upon result publication via configured notification channels.

7.6 Accounting and Finance

The Accounting and Finance module provides comprehensive financial management covering fee collection, payroll processing, and financial reporting, integrated with payment gateways for online transactions.

7.6.1 Fee Management

FR-FEE-01	Fee Structure Configuration	Define fee heads, amounts, due dates, and discounts per campus, program, and term.
FR-FEE-02	Automated Invoice Generation	System generates fee invoices for all enrolled students at each billing cycle without manual intervention.
FR-FEE-03	Multi-Channel Payment Collection	Accept payments via bank transfer, online payment gateway, and cash with receipt confirmation.
FR-FEE-04	Digital Receipt Issuance	Automatic digital receipt generation and delivery to portal upon payment confirmation.
FR-FEE-05	Defaulter Management	Automated overdue account identification with configurable escalation: reminder → late fee → access restriction.

FR-FEE-06	Concession and Scholarship Management	Manage merit-based, sibling, and need-based concessions with approval workflows and financial impact reporting.
FR-FEE-07	Fee Refund Processing	Structured refund workflow with approval routing, adjustment posting, and bank transfer initiation.
FR-FEE-08	Fee Collection Analytics	Daily, monthly, and annual dashboards by campus, program, and payment method. Outstanding liability reporting.

7.6.2 Payroll Management

FR-PAY-01	Employee Payroll Setup	Configure salary components (basic, allowances, deductions, EOBI, income tax) per employee grade.
FR-PAY-02	Monthly Payroll Processing	Automated payroll computation incorporating attendance deductions, leave adjustments, and incremental changes.
FR-PAY-03	Payslip Generation	Digital payslip generation and delivery via employee portal upon payroll approval.
FR-PAY-04	Payroll Approval Workflow	HR prepares → Finance reviews → Principal/Director approves → Disbursement authorized.
FR-PAY-05	Tax and Statutory Compliance	Automated computation of income tax deductions and EOBI contributions per regulatory schedules.
FR-PAY-06	Payroll Analytics	Cost center-wise payroll analysis, headcount cost trends, and year-over-year salary expenditure reporting.

7.7 Human Resource Management

The HRM module manages the complete employee lifecycle from recruitment through offboarding, supporting all staff categories across all 31 campuses.

FR-HRM-01	Employee Master Records	Comprehensive digital personnel files: personal data, qualifications, employment history, and contracts.
FR-HRM-02	Recruitment Pipeline	Post vacancies, manage applications, schedule interviews, issue offer letters, and onboard new hires.
FR-HRM-03	Contract Management	Track employment contracts, probation periods, confirmation dates, and renewals with expiry alerts.
FR-HRM-04	Leave Management	Digital leave application, balance tracking, and multi-level approval. Support for annual, casual, medical, and maternity leave types.
FR-HRM-05	Staff Attendance Tracking	Daily attendance recording with late/early departure flagging and payroll integration for deductions.
FR-HRM-06	Performance Management	Annual appraisal workflows with configurable criteria, self-assessment, and principal/HOD review.
FR-HRM-07	Training and Development	Log completed training programs, certifications, and professional development per employee.
FR-HRM-08	Disciplinary Record Management	Document warnings, show-cause notices, and disciplinary actions with full audit trail.
FR-HRM-09	Separation Management	Process resignations, terminations, and retirements with clearance checklists and final settlement.
FR-HRM-10	Organizational Chart	Live auto-generated organizational chart reflecting current staff assignments by campus and department.
FR-HRM-11	HR Analytics	Headcount reports, turnover analysis, leave utilization rates, and staff qualification distribution.

7.8 Assets and Inventory Management

The Assets and Inventory module provides complete visibility and control over institutional fixed assets and consumable inventory across all campuses.

FR-AST-01	Asset Registry	Centralized register of all fixed assets with unique tags, location, custodian assignment, and acquisition details.
FR-AST-02	Asset Lifecycle Management	Track asset status: Active, Under Maintenance, Condemned, Disposed. Record maintenance events and costs.
FR-AST-03	Depreciation Tracking	Configurable straight-line and reducing balance depreciation schedules with automated annual posting.
FR-AST-04	Asset Allocation and Transfer	Workflow for allocating assets to staff/departments and transferring between campuses with custody chain documentation.
FR-AST-05	Inventory Stock Management	Real-time stock levels for consumables with configurable reorder point alerts triggering procurement requisitions.

FR-AST-06	Stock Issue and Return	Digital stock issue vouchers and return receipts with requisitioning department attribution.
FR-AST-07	Annual Asset Verification	System-guided verification workflow with physical count reconciliation and variance reporting.
FR-AST-08	Disposal Management	Structured write-off and disposal workflow with approval routing and auction/scrap record maintenance.
FR-AST-09	Asset Analytics	Asset utilization reports, depreciation schedules, maintenance cost analysis, and campus-wise valuation.

7.9 Cross-Cutting Platform Features

The following features are embedded throughout the FAMS platform, providing consistent enterprise-grade capabilities across all functional modules.

FR-PLT-01	Real-Time Analytics Dashboards	Executive and module-specific dashboards with live KPI tiles, trend charts (line, bar, pie, heat map), and exportable reports in PDF and Excel.
FR-PLT-02	AI Chatbot	NLP-based conversational assistant resolving Tier-1 queries (fee balances, exam schedules, results, timetables) with escalation routing.
FR-PLT-03	Notifications Engine	Event-driven system delivering SMS, email, and in-app alerts for 40+ defined trigger events across all modules.
FR-PLT-04	E-Learning Integration	SSO bridge to external LMS. Synchronize enrollment data, track completion, and surface progress within the FAMS student portal.
FR-PLT-05	Document Generation Engine	Template-driven generation of grade cards, fee receipts, offer letters, payslips, purchase orders, and asset labels.
FR-PLT-06	Multi-Campus Configuration	Campus-level configuration of academic calendars, fee structures, grading scales, and organizational hierarchies.
FR-PLT-07	Audit Trail and Activity Logging	Immutable log of all create, update, delete, and access events across every module, user, and campus.
FR-PLT-08	Role-Based Access Control	Granular permission management at module, feature, and data-record levels. All access changes require administrator authorization.
FR-PLT-09	Bulk Operations	Bulk data import, export, and update capabilities across all modules for high-volume administrative workflows.
FR-PLT-10	Offline-Capable Attendance	Attendance data captured on tablets queued locally when offline and synchronized automatically upon reconnection.

8. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes that FAMS must exhibit in addition to its functional capabilities. These requirements govern performance, reliability, security, maintainability, and usability and are binding on the development and infrastructure teams.

NFR-01	Performance	Web portal response time	Page load under 2 s (10 Mbps). API 95th-percentile response under 500 ms.
NFR-02	Performance	Concurrent user load	Minimum 5,000 concurrent authenticated sessions without performance degradation.
NFR-03	Availability	Uptime SLA	99.9% availability — maximum 8.7 hours unplanned downtime per year.
NFR-04	Availability	Planned maintenance	Scheduled windows ≤ 4 hours/month, outside peak hours (06:00–22:00 PKT).
NFR-05	Reliability	Data backup	Automated daily backups. 90-day retention. RTO: 4 hours. RPO: 24 hours.
NFR-06	Reliability	Failover	Automatic failover to secondary infrastructure; service restoration within 15 minutes.
NFR-07	Scalability	Horizontal scaling	Application tier auto-scales to accommodate peak loads up to 3× baseline.
NFR-08	Scalability	Data growth	Database supports growth to 10 million records without architectural changes.
NFR-09	Security	Authentication	MFA mandatory for all administrator and principal accounts. Password complexity enforced.
NFR-10	Security	Data encryption	AES-256 at rest. TLS 1.3 in transit. No plaintext PII transmission.
NFR-11	Security	Session management	Idle timeout: 30 min (staff), 60 min (student/parent). Re-auth for sensitive operations.
NFR-12	Usability	Browser compatibility	Full support on latest 2 versions of Chrome, Firefox, Safari, Edge. Responsive ≥ 768px.
NFR-13	Usability	Accessibility	WCAG 2.1 Level AA compliance for all portal pages.
NFR-14	Maintainability	Code quality	Minimum 80% unit test coverage. CI/CD with zero-downtime deployments for patches.
NFR-15	Compliance	Data privacy	Compliance with applicable Pakistani data protection regulations.
NFR-16	Audit	Log retention	All system and user activity logs retained for minimum 5 years in tamper-evident storage.

Table II. Non-Functional Requirements

9. SYSTEM ARCHITECTURE

FAMS is built on a modern multi-tier, microservices-influenced architecture designed for cloud deployment. The architecture prioritizes separation of concerns, independent deployability of modules, horizontal scalability, and data security.

9.1 Presentation Layer

The presentation layer consists of a single-page web application (SPA) built using React.js, served via a global CDN ensuring fast load times for all campus locations. The UI is fully responsive, supporting desktop, laptop, and tablet viewports. The tablet attendance interface is developed as a progressive web application (PWA) to support offline operation and local data queuing. Role-based rendering ensures each authenticated user sees only modules and data elements relevant to their role. All API calls are authenticated via JSON Web Tokens (JWT) with short expiry windows and refresh token rotation.

9.2 API Gateway Layer

All client requests pass through a centralized API Gateway handling authentication token validation, rate limiting, request routing to downstream microservices, and response aggregation. The gateway exposes a versioned RESTful API (v1, v2, etc.) to maintain backward compatibility across client upgrades and provides a single ingress point for all API traffic, simplifying security enforcement and traffic monitoring.

9.3 Application Layer

The application layer is decomposed into domain-aligned service groups corresponding to the eight FAMS functional modules plus shared services (notifications, document generation, AI chatbot, analytics). Each service group is independently deployable and horizontally scalable. Services communicate synchronously via HTTP/REST for user-facing operations and asynchronously via a message broker (Redis Streams or RabbitMQ) for event-driven workflows such as notification dispatch and audit logging.

9.4 Data Layer

The primary data store is a PostgreSQL relational database hosted on a managed cloud service with automated replication, failover, and point-in-time recovery. Multi-campus data isolation is enforced through row-level security (RLS) with campus_id as the discriminator column across all tenant-scoped tables. A Redis cache layer handles session storage and high-frequency read operations. A cloud object store (S3-compatible) holds all binary assets including uploaded documents, generated PDFs, and report exports.

9.5 Infrastructure Layer

The platform is deployed on a major cloud provider (AWS or Microsoft Azure) in a multi-availability-zone configuration. Application containers are orchestrated using Kubernetes with horizontal pod autoscaling (HPA) tied to CPU utilization and request queue depth metrics. Infrastructure-as-code (Terraform) is used for all environment provisioning. All external traffic is routed through a Web Application Firewall (WAF) providing protection against OWASP Top 10 vulnerabilities. DDoS mitigation is provided at the CDN and load balancer layers.

10. DATA FLOW AND INTEGRATION

10.1 Internal Data Flows

Within FAMS, data flows between modules follow defined event patterns. Table III summarizes the principal cross-module data flows.

DF-INT-01	Admissions → CRM	Enrollment confirmation triggers creation of full student profile and portal credential provisioning.
DF-INT-02	Admissions → Finance	Enrollment confirmation triggers student fee account creation and first-term invoice generation.
DF-INT-03	Attendance → Results	Attendance records feed examination eligibility computation, enforcing minimum attendance thresholds.
DF-INT-04	Attendance → Finance	Approved leave and attendance data feed payroll for leave-without-pay deductions.
DF-INT-05	HR → Finance	Employee profile data from HRM feeds the payroll processing engine.
DF-INT-06	Procurement → Finance	Approved vendor invoices generate financial liability entries and trigger payment authorization.
DF-INT-07	Procurement → Assets	Goods receipt confirmations for capital assets trigger asset registration entries.
DF-INT-08	Results → CRM	Published result data appended to student CRM profile for longitudinal academic history.
DF-INT-09	All Modules → Notify	Defined system events trigger the Notifications Engine to dispatch SMS, email, or in-app alerts.
DF-INT-10	All Modules → Analytics	All transactional events stream to the Analytics layer for real-time dashboard updates.

Table III. Internal Data Flows

10.2 External Integrations

INT-EXT-01	Payment Gateway	REST API	Bi-directional — FAMS sends payment requests and receives real-time confirmation webhooks.
INT-EXT-02	SMS Gateway	REST API	Outbound — Transactional SMS notifications. Delivery receipts logged.
INT-EXT-03	Email Service	SMTP / API	Outbound — Transactional emails: receipts, result notifications, offer letters, payslips.
INT-EXT-04	Learning Management System	REST API + SSO	Bi-directional — FAMS pushes enrollment/class data; pulls e-learning completion status.
INT-EXT-05	Exam Board (Phase 2)	REST API	Outbound — Registration and result data submission to external examination bodies.

INT-EXT-06	Biometric Devices	Device SDK	Inbound — Optional integration for staff biometric check-in/check-out data ingestion.

Table IV. External System Integrations

11. SECURITY AND COMPLIANCE

Security is treated as a first-class system attribute in FAMS. The platform implements a defense-in-depth security model spanning identity management, data protection, network security, and application-level controls, designed to protect the personal data of students, parents, and staff in compliance with applicable regulations.

11.1 Identity and Access Management

Multi-factor authentication (MFA) is mandatory for all administrative roles. Single sign-on (SSO) is supported via SAML 2.0 for institutional identity provider integration. All accounts are subject to the principle of least privilege enforced at provisioning. Dormant accounts are automatically suspended after 90 days of inactivity.

11.2 Data Encryption

AES-256 encryption is applied for all data at rest. TLS 1.3 is enforced for all data in transit between clients, the API gateway, and backend services. Database backups are encrypted before storage. Encryption keys are managed via a cloud-native KMS with automatic rotation policies.

11.3 Application Security

The platform is developed in compliance with the OWASP Application Security Verification Standard (ASVS) Level 2. Automated SAST and DAST scans are integrated into the CI/CD pipeline. Independent third-party penetration testing is conducted before production deployment and annually thereafter.

11.4 Network Security

All inbound public traffic is routed through a WAF. Backend services are deployed in private subnets with no direct public internet exposure. Security group rules enforce deny-by-default ingress policies. DDoS protection is active at CDN and load balancer tiers.

11.5 Audit and Logging

An immutable audit log records every authenticated user action — data read, create, update, and delete events — with timestamps, user identity, IP address, and affected record identifiers. Logs are stored in tamper-evident storage with a minimum 5-year retention. Anomalous access patterns trigger automated security alerts.

11.6 Data Privacy and Compliance

FAMS complies with applicable Pakistani data protection frameworks. PII is pseudonymized in non-production environments. Data subject access and deletion request workflows are provided. Data sharing with third parties is governed by explicit data processing agreements.

12. SCALABILITY CONSIDERATIONS

FAMS is architected to scale both horizontally (additional compute nodes) and vertically (increased node capacity) in response to institutional growth and demand variability.

12.1 Campus Growth

The multi-tenant architecture supports the addition of new campuses without schema changes or application redeployment. A new campus is provisioned through an administrative configuration workflow. The system is certified to support up to 100 campuses under the current architecture without performance impact.

12.2 User Volume Growth

HPA policies automatically provision additional application instances when CPU utilization or request queue depth exceeds defined thresholds. Load testing benchmarks must validate the system at 3× projected peak concurrent user volume before go-live.

12.3 Data Volume Growth

PostgreSQL is deployed with read replicas to distribute query load. Partitioning by campus_id and academic year is implemented for high-volume tables. Archive policies move records older than 7 years to cold storage while maintaining query accessibility.

12.4 Module Extensibility

The microservices architecture allows new functional modules to be developed and deployed independently without impacting existing modules. The API Gateway routing configuration and shared authentication service ensure new modules are immediately accessible upon deployment.

12.5 Integration Scalability

The event-driven messaging architecture ensures increases in event volume do not create cascading latency in notification and analytics pipelines. Message broker capacity scales independently of the application tier.

13. RISKS AND ASSUMPTIONS

13.1 Project Assumptions

- Falcon College will designate a dedicated Product Owner and steering committee with authority to make product decisions within agreed timelines.
- All 31 campuses have adequate internet connectivity (minimum 10 Mbps dedicated broadband) to support web application access and real-time synchronization.
- The client will provide access to existing data in structured format for migration and assign a data owner for each module.
- Tablet hardware for the attendance system will be procured, configured, and deployed by the client's IT team to RaideIT's hardware specification.
- The external LMS for e-learning integration has a publicly documented REST API and will be provisioned before the integration sprint begins.
- Content for the AI chatbot knowledge base (FAQs, institutional policies, process guides) will be provided by the client in structured format.
- The client accepts that UX design will follow RaideIT's established design system; custom branding is limited to logo, color scheme, and typography.

13.2 Risk Register

R-01	Data Migration Complexity	High	High	Legacy data in inconsistent formats may extend migration timelines.	Data audit in discovery phase. Define acceptance criteria. Dedicated migration sprint with rollback plan.
R-02	Campus Connectivity Variance	High	Medium	Inadequate connectivity at some campuses may degrade application performance.	Connectivity audit pre-deployment. Offline PWA for attendance. Minimum connectivity standard in infrastructure SLA.
R-03	Stakeholder Adoption Resistance	Medium	High	Staff accustomed to legacy processes may resist platform adoption.	Phased rollout with pilot campuses. Structured change management and end-user training programs.
R-04	Scope Creep	High	Medium	Incremental feature additions may inflate scope and delay delivery.	Strict change control. All scope changes require formal CR with impact assessment and steering committee approval.
R-05	Third-Party API Instability	Medium	Low	Payment gateway or SMS provider changes or outages may disrupt integrated services.	Select tier-1 providers with SLAs. Implement graceful degradation. Maintain fallback provider configurations.
R-06	Regulatory Change	Low	High	Changes to Pakistani data protection legislation may require platform modifications.	Privacy-by-design architecture. Monitor regulatory developments. Annual compliance review.

R-07	Key Personnel Dependency	Medium	Medium	Dependency on specific team members may create delivery risk.	Enforce documentation standards and knowledge sharing. Cross-train team members on critical modules.

Table V. Risk Register

14. CONCLUSION

This Product Requirements Document represents the complete, authorized specification for the Falcon Academic Management System (FAMS), an enterprise ERP platform designed to transform the operational infrastructure of Falcon College's 31-campus network. The document has been prepared by RaideIT Software Solutions as the principal delivery partner and reflects requirements gathered through structured discovery sessions with Falcon College's leadership, administrative, and academic teams.

FAMS addresses a critical institutional need: consolidation of fragmented, campus-level administrative operations into a unified, centrally governed digital ecosystem. By delivering a single platform spanning CRM, Admissions, Procurement, Academic Operations, Results and Reporting, Finance, HRM, and Asset Management, FAMS enables Falcon College to operate with greater efficiency, data accuracy, and service quality across every campus and every stakeholder touchpoint.

The platform's commitment to real-time data synchronization, AI-augmented support, tablet-based operational workflows, and fully paperless transactional processes positions Falcon College at the forefront of institutional digital maturity. The architecture is designed to accommodate future technology investments without architectural debt.

Upon stakeholder approval of this document, RaideIT will proceed to the technical design phase, producing detailed system design specifications, API contracts, and sprint planning artifacts aligned with the agreed delivery roadmap. All subsequent development, testing, and deployment activities will be governed by this PRD as the authoritative reference for scope, requirements, and acceptance criteria.

Prepared By	Reviewed By	Approved By
RaideIT Software Solutions and Mehboob.mov. Product & Engineering Team	Falcon College Steering Committee	Falcon College Board of Governors
______________________	______________________	______________________
Signature & Date	*Signature & Date*	*Signature & Date*